FMCG 4.0 Formal Verification: Complete Theorem Reference

Lean 4 / Mathlib Formalisation — 14 Machine-Checked Theorems

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Abstract

This document presents the complete set of fourteen machine-checked theorems formalising the mathematical properties of the Signal-to-Shelf (StS) metric and Coupling Benefit Ratio (CBR), as defined in the FMCG 4.0 framework. All proofs were developed and verified in Lean 4 using the Mathlib library, with zero admitted proofs (`sorry`) and zero unresolved proof obligations. The complete formalisation is publicly available at: https://github.com/Alex-Alves3/FMCG40-Lean

Core Definitions

Signal-to-Shelf (StS)

$$\mathrm{StS}(R_0, \lambda, t_0, t) = \frac{R_0 \cdot e^{-\lambda t}}{t + t_0} \tag{1}$$

Market Response

$$\mathrm{MarketResponse}(R_0, \lambda, t) = R_0 \cdot e^{-\lambda t} \tag{2}$$

Temporal Decay

$$D(\lambda, t) = e^{-\lambda t} \tag{3}$$

Integrated / Sequential Intrinsic Quality

$$R_{0,\text{integrated}} = R_{\text{base}} + R_{\text{prod}} + R_{\text{mark}} + R_{\text{coup}} \tag{4}$$

$$R_{0,\text{sequential}} = R_{\text{base}} + R_{\text{prod}} + R_{\text{mark}} \tag{5}$$

Coupling Benefit Ratio (CBR)

$$\mathrm{CBR} = \frac{\mathrm{StS}(R_{0,\text{integrated}}, \lambda, t_0, t_{\text{int}})}{\mathrm{StS}(R_{0,\text{sequential}}, \lambda, t_0, t_{\text{seq}})} \tag{6}$$

Coupling Contribution Percentage

$$\eta_{\text{coup}} = \frac{R_{\text{coup}}}{R_{0,\text{integrated}}} \tag{7}$$

1 StS Denominator Positivity

Statement. For $t_0 > 0$ and $t \geq 0$, the StS denominator is strictly positive.

$$t_0 > 0,\ t \geq 0 \implies t + t_0 > 0 \tag{8}$$

```
theorem StS_denominator_pos (t0 t : R) (ht0 : t0 > 0) (
    ht : t >= 0) :
t + t0 > 0 := by
linarith
```

Significance. The foundational well-definedness fact — establishes that StS never divides by zero under the stated parameter conditions.

2 StS Well-Definedness

Statement. The StS denominator is never exactly zero.

$$t_0 > 0,\ t \geq 0 \implies t + t_0 \neq 0 \tag{9}$$

```
theorem StS_well_defined (t0 t : R) (ht0 : t0 > 0) (ht :
    t >= 0) :
t + t0 <> 0 := by
have h := StS_denominator_pos t0 t ht0 ht
linarith
```

Significance. Direct corollary of Theorem 1, stated in the form needed for division-safety arguments elsewhere.

3 Maximum StS Value at $t = 0$

Statement. At $t = 0$, StS reduces exactly to R_0/t_0.

$$\mathrm{StS}(R_0, \lambda, t_0, 0) = \frac{R_0}{t_0} \tag{10}$$

```
theorem StS_max_value (R0 lam t0 : R) (ht0 : t0 > 0) :
StS R0 lam t0 0 = R0 / t0 := by
simp [StS, Real.exp_zero]
```

Significance. Confirms the peak efficiency value the paper's prose asserts — immediate deployment yields exactly R_0/t_0, with no decay term reducing it.

4 Market Response Positivity

Statement. For $R_0 > 0$, the Market Response function is strictly positive at every t.

$$R_0 > 0 \implies R_0 \cdot e^{-\lambda t} > 0 \tag{11}$$

```
theorem MarketResponse_pos (R0 lam t : R) (hR0_pos : R0
    > 0) :
MarketResponse R0 lam t > 0 := by
unfold MarketResponse
apply mul_pos hR0_pos
exact Real.exp_pos _
```

Significance. Establishes that market response never becomes zero or negative for a strategy with positive intrinsic quality — a basic sanity property of the model.

5 Temporal Decay Bounds

Statement. For $t \geq 0$ and $\lambda > 0$, the decay function stays strictly between 0 and 1.

$$0 < e^{-\lambda t} \leq 1 \tag{12}$$

```
theorem TemporalDecay_bounds (lam t : R) (ht : t >= 0) (
    hlam : lam > 0) :
0 < TemporalDecay lam t /\ TemporalDecay lam t <= 1 :=
    by
constructor
. unfold TemporalDecay; exact Real.exp_pos _
. unfold TemporalDecay
rw [<- Real.exp_zero]
apply Real.exp_le_exp.mpr
nlinarith
```

Significance. Confirms the decay function behaves as a genuine "fraction of opportunity remaining" — never exceeding 100%, never reaching or going below 0%.

6 Trend Half-Life Identity

Statement. The formula $t_{1/2} = \ln(2)/\lambda$ correctly gives the point at which decay reaches exactly one-half.

$$D\left(\lambda, \frac{\ln 2}{\lambda}\right) = \frac{1}{2} \tag{13}$$

```
theorem trend_half_life (lam : R) (hlam : lam > 0) :
TemporalDecay lam (Real.log 2 / lam) = 1 / 2 := by
unfold TemporalDecay
have hne : lam <> 0 := ne_of_gt hlam
rw [show -lam * (Real.log 2 / lam) = -Real.log 2 by
    field_simp]
rw [Real.exp_neg, Real.exp_log (by norm_num : (2:R) > 0)
    ]
norm_num
```

Significance. Confirms the half-life formula used throughout the paper's parameter estimation discussion is exact, not approximate.

7 Uniqueness of Exponential Decay

Statement. Given $D(0) = 1$ and the derivative condition $D'(t) = -\lambda D(t)$ at every $t \geq 0$, the function D is uniquely determined to be $D(t) = e^{-\lambda t}$.

$$D(0) = 1,\ \forall t \geq 0,\ D'(t) = -\lambda D(t) \implies \forall t \geq 0,\ D(t) = e^{-\lambda t} \tag{14}$$

```
theorem decay_uniqueness_given_derivative_at_zero
(D : R -> R)
(hD0 : D 0 = 1)
(hDpos : forall t >= 0, D t > 0)
```

```
(hDfunc : forall t s, t >= 0 -> s >= 0 -> D (t + s) = D
   t * D s)
(lam : R)
(hDderiv_all : forall t >= 0, HasDerivAt D (-lam * D t)
   t) :
forall t >= 0, D t = Real.exp (-lam * t) := by
intro t ht
set E : R -> R := fun s => D s * Real.exp (lam * s) with
    hE_def
have hEderiv_pt : forall s >= 0, HasDerivAt E 0 s := by
intro s hs0
have hs1 : HasDerivAt (fun u : R => Real.exp (lam * u))
   (Real.exp (lam * s) * lam) s := by
have hlin : HasDerivAt (fun u : R => lam * u) lam s :=
   by
simpa using (hasDerivAt_id s).const_mul lam
simpa [mul_comm] using hlin.exp
have h2 := (hDderiv_all s hs0).mul hs1
have heq : -lam * D s * Real.exp (lam * s) + D s * (Real
   .exp (lam * s) * lam) = 0 := by
ring
rw [<- heq]
exact h2
have hEderiv_within : forall s in Set.Ico (0:R) t,
   HasDerivWithinAt E 0 (Set.Ici s) s :=
fun s hs => (hEderiv_pt s hs.1).hasDerivWithinAt
have hEcont : ContinuousOn E (Set.Icc 0 t) :=
HasDerivAt.continuousOn (fun x hx => hEderiv_pt x hx.1)
have hbound : forall x in Set.Ico (0:R) t, ||(0:R)|| <=
   (0:R) := by
intro x _; simp
have hkey :=
   norm_image_sub_le_of_norm_deriv_right_le_segment
(f := E) (f' := fun _ => (0:R)) (a := (0:R)) (b := t) (C
    := (0:R))
hEcont hEderiv_within hbound
have hEt : E t = E 0 := by
have h0 : ||E t - E 0|| <= 0 * (t - 0) := hkey t (Set.
   right_mem_Icc.mpr ht)
have h0' : E t - E 0 = 0 := by
have h1 : ||E t - E 0|| <= 0 := by linarith
exact norm_le_zero_iff.mp h1
linarith [h0']
have hE0 : E 0 = 1 := by simp [hE_def, hD0]
have hEt1 : E t = 1 := by rw [hEt, hE0]
have hkey2 : D t * Real.exp (lam * t) = 1 := hEt1
have hExp_pos : Real.exp (lam * t) > 0 := Real.exp_pos _
have htarget : D t * Real.exp (lam * t) = Real.exp (-lam
    * t) * Real.exp (lam * t) := by
rw [hkey2]
rw [<- Real.exp_add]
have : -lam * t + lam * t = 0 := by ring
rw [this, Real.exp_zero]
exact mul_right_cancel0 (ne_of_gt hExp_pos) htarget
```

Significance. Closes the gap between the paper's prose claim ("the unique continuous function satisfying the memoryless property") and formal verification. This proof establishes uniqueness under an *explicit differentiability hypothesis* — a genuinely honest, if narrower, claim than the

fully general classical theorem (which derives differentiability from continuity plus the functional equation alone, and remains unformalised here). Proof strategy: construct the auxiliary function $E(s) = D(s)e^{\lambda s}$, show it has zero derivative everywhere on $[0, t]$, invoke a mean-value-theorem-based constancy result to conclude E is constant on the interval, then solve algebraically for $D(t)$.

8 Coupling Contribution Bound

Statement. Given $R_{\text{coup}} \geq 0$ and $R_{\text{base}} + R_{\text{prod}} + R_{\text{mark}} > 0$, the coupling contribution percentage lies strictly in $[0, 1)$.

$$0 \leq \eta_{\text{coup}} < 1 \tag{15}$$

```
theorem eta_coup_bounds
(R_base R_prod R_mark R_coup : R)
(hR_coup : R_coup >= 0)
(hR_base_pos : R_base + R_prod + R_mark > 0) :
0 <= eta_coup R_base R_prod R_mark R_coup /\
eta_coup R_base R_prod R_mark R_coup < 1 := by
have hden_pos : R0_integrated R_base R_prod R_mark
   R_coup > 0 := by
unfold R0_integrated
linarith
unfold eta_coup
constructor
. exact div_nonneg hR_coup (le_of_lt hden_pos)
. rw [div_lt_one hden_pos]
unfold R0_integrated
linarith
```

Significance. Confirms η_{coup} behaves as a valid percentage — it can never exceed 100% of the total strategy quality, and is never negative.

9 Decay Rate Estimator Positivity

Statement. Given two trend strength measurements $TS_1 > TS_2 > 0$ at times $t_1 < t_2$, the estimated decay rate is positive.

$$TS_1 > TS_2 > 0,\ t_1 < t_2 \implies \frac{\ln(TS_1/TS_2)}{t_2 - t_1} > 0 \tag{16}$$

```
theorem decay_rate_estimator_well_defined
(TS1 TS2 t1 t2 : R)
(hTS1 : TS1 > 0) (hTS2 : TS2 > 0)
(hTS_decay : TS1 > TS2)
(ht_order : t1 < t2) :
Real.log (TS1 / TS2) / (t2 - t1) > 0 := by
apply div_pos
. apply Real.log_pos
rw [lt_div_iff0 hTS2]
linarith
. linarith
```

Significance. Validates the parameter estimation procedure described in the paper — a decaying trend (measured strength decreasing over time) always produces a sensible, positive λ estimate.

10 Strict Monotonicity of StS

Statement. For $R_0 > 0$, $\lambda > 0$, $t_0 > 0$, the StS function is strictly decreasing on $[0, \infty)$.

$$\mathrm{StS}(R_0, \lambda, t_0, \cdot) \text{ is strictly decreasing on } [0, \infty) \tag{17}$$

```
theorem StS_strictly_decreasing
(R0 lam t0 : R) (hR0_pos : R0 > 0) (hlam : lam > 0) (ht0
    : t0 > 0) :
StrictAntiOn (StS R0 lam t0) (Ici 0) := by
have hderiv : forall t in Ici (0:R),
HasDerivAt (StS R0 lam t0)
(R0 * (Real.exp (-lam * t) * (-lam * (t + t0) - 1)) / (t
    + t0)^2) t := by
intro t ht
have ht' : t >= 0 := ht
have hnum : HasDerivAt (fun s : R => R0 * Real.exp (-lam
    * s))
(R0 * (Real.exp (-lam * t) * (-lam))) t := by
have h1 : HasDerivAt (fun s : R => -lam * s) (-lam) t :=
    by
simpa using (hasDerivAt_id t).const_mul (-lam)
have h2 : HasDerivAt (fun s : R => Real.exp (-lam * s))
(Real.exp (-lam * t) * (-lam)) t := h1.exp
simpa [mul_assoc] using h2.const_mul R0
have hden : HasDerivAt (fun s : R => s + t0) 1 t := by
simpa using (hasDerivAt_id t).add_const t0
have hdenne : (t + t0) <> 0 := ne_of_gt (
   StS_denominator_pos t0 t ht0 ht')
have hcomb := hnum.div hden hdenne
have hfun : StS R0 lam t0 = (fun s : R => R0 * Real.exp
   (-lam * s)) / (fun s : R => s + t0) := by
funext s
simp [StS, Pi.div_apply]
rw [hfun]
have heq : R0 * (Real.exp (-lam * t) * (-lam * (t + t0)
   - 1)) / (t + t0) ^ 2
= (R0 * (Real.exp (-lam * t) * (-lam)) * (t + t0) - R0 *
    Real.exp (-lam * t) * 1)
/ (t + t0) ^ 2 := by ring
rw [heq]
exact hcomb
apply strictAntiOn_of_deriv_neg (convex_Ici (0:R))
. apply ContinuousOn.div
. exact (continuous_const.mul (Real.continuous_exp.comp
(continuous_const.mul continuous_id))).continuousOn
. exact (continuous_id.add continuous_const).
   continuousOn
. intro t ht
exact ne_of_gt (StS_denominator_pos t0 t ht0 ht)
. intro t ht
rw [interior_Ici] at ht
have ht' : t >= 0 := le_of_lt ht
have hd := (hderiv t ht').deriv
rw [hd]
have hexp_pos : Real.exp (-lam * t) > 0 := Real.exp_pos
   _
have hden_pos : (t + t0)^2 > 0 := by positivity
```

```
    have hfactor_neg : (-lam * (t + t0) - 1) < 0 := by
    have hpos : lam * (t + t0) > 0 := mul_pos hlam (by
        linarith)
    linarith
    have hnum_neg : R0 * (Real.exp (-lam * t) * (-lam * (t +
         t0) - 1)) < 0 := by
    apply mul_neg_of_pos_of_neg hR0_pos
    exact mul_neg_of_pos_of_neg hexp_pos hfactor_neg
    exact div_neg_of_neg_of_pos hnum_neg hden_pos
```

Significance. The formal justification for "acting faster is always better" — proved via direct computation of the derivative and showing it is negative everywhere. This is the mathematical backbone of the paper's central speed argument.

11 Maximum StS at $t = 0$ (Global)

Statement. The maximum value of StS over all $t \geq 0$ occurs at $t = 0$.

$$\forall t \geq 0,\ \mathrm{StS}(R_0, \lambda, t_0, t) \leq \mathrm{StS}(R_0, \lambda, t_0, 0) \tag{18}$$

```
    theorem StS_max_at_zero
    (R0 lam t0 t : R)
    (hR0_pos : R0 > 0) (hlam : lam > 0) (ht0 : t0 > 0) (ht :
         t >= 0) :
    StS R0 lam t0 t <= StS R0 lam t0 0 := by
    have hanti : AntitoneOn (StS R0 lam t0) (Ici 0) :=
    (StS_strictly_decreasing R0 lam t0 hR0_pos hlam ht0).
        antitoneOn
    exact hanti (mem_Ici.mpr (le_refl 0)) (mem_Ici.mpr ht)
        ht
```

Significance. Direct consequence of Theorem 10 — confirms the maximum achievable efficiency R_0/t_0 (Theorem 3) is genuinely the global maximum, not merely a local one.

12 Sufficient Conditions for CBR ≥ 1

Statement. Given $t_{\text{int}} \geq 0$, $t_{\text{seq}} \geq 0$, $t_{\text{int}} \leq t_{\text{seq}}$, $R_{\text{coup}} > 0$, and $R_{\text{base}} + R_{\text{prod}} + R_{\text{mark}} > 0$, the Coupling Benefit Ratio is at least 1.

$$\mathrm{CBR} \geq 1 \tag{19}$$

```
    theorem CBR_geq_one
    (R_base R_prod R_mark R_coup lam t0 t_int t_seq : R)
    (hlam : lam > 0) (ht0 : t0 > 0)
    (ht_int : t_int >= 0)
    (ht_seq : t_seq >= 0)
    (ht : t_int <= t_seq)
    (hcoup : R_coup > 0)
    (hR_base_pos : R_base + R_prod + R_mark > 0) :
    CBR R_base R_prod R_mark R_coup lam t0 t_int t_seq >= 1
        := by
    set R0int := R_base + R_prod + R_mark + R_coup with
        hR0int_def
    set R0seq := R_base + R_prod + R_mark with hR0seq_def
    have hR0int_pos : R0int > 0 := by rw [hR0int_def];
        linarith
```

```
    have hR0seq_pos : R0seq > 0 := by rw [hR0seq_def];
       linarith
    have hR0_lt : R0seq < R0int := by rw [hR0int_def,
       hR0seq_def]; linarith
    have hden_int_pos : t_int + t0 > 0 :=
       StS_denominator_pos t0 t_int ht0 ht_int
    have hden_seq_pos : t_seq + t0 > 0 :=
       StS_denominator_pos t0 t_seq ht0 ht_seq
    have hStS_seq_pos : StS R0seq lam t0 t_seq > 0 := by
    unfold StS
    exact div_pos (mul_pos hR0seq_pos (Real.exp_pos _))
       hden_seq_pos
    have step_a : StS R0seq lam t0 t_seq <= StS R0int lam t0
        t_seq := by
    unfold StS
    apply div_le_div_of_nonneg_right _ (le_of_lt
       hden_seq_pos)
    exact mul_le_mul_of_nonneg_right (le_of_lt hR0_lt) (
       le_of_lt (Real.exp_pos _))
    have step_b : StS R0int lam t0 t_seq <= StS R0int lam t0
        t_int := by
    rcases eq_or_lt_of_le ht with heq | hlt
    . rw [heq]
    . have hanti : AntitoneOn (StS R0int lam t0) (Ici 0) :=
    (StS_strictly_decreasing R0int lam t0 hR0int_pos hlam
       ht0).antitoneOn
    exact hanti (mem_Ici.mpr ht_int) (mem_Ici.mpr ht_seq) ht
    unfold CBR
    rw [ge_iff_le, le_div_iff0 hStS_seq_pos]
    calc 1 * StS R0seq lam t0 t_seq = StS R0seq lam t0 t_seq
        := by ring
    _ <= StS R0int lam t0 t_seq := step_a
    _ <= StS R0int lam t0 t_int := step_b
```

Significance. The paper's central claim, now genuinely proved rather than merely asserted — when integration is at least as fast as the sequential alternative AND provides positive coupling value, integration is mathematically guaranteed to be at least as efficient. Proof combines Theorem 10 (monotonicity in time) with a separate monotonicity-in-R_0 argument.

13 CBR Invariance to λ (Equal Deployment Times)

Statement. When $t_{\text{int}} = t_{\text{seq}} = t$, the Coupling Benefit Ratio is exactly invariant to the choice of λ.

$$t_{\text{int}} = t_{\text{seq}} \implies \text{CBR}(\lambda_1) = \text{CBR}(\lambda_2)\ \forall \lambda_1, \lambda_2 > 0 \tag{20}$$

```
    theorem CBR_lambda_invariant_equal_time
    (R_base R_prod R_mark R_coup t0 t lam1 lam2 : R)
    (ht0 : t0 > 0) (ht : t >= 0)
    (hlam1 : lam1 > 0) (hlam2 : lam2 > 0)
    (hR_base_pos : R_base + R_prod + R_mark > 0)
    (hcoup : R_coup > 0) :
    CBR R_base R_prod R_mark R_coup lam1 t0 t t =
    CBR R_base R_prod R_mark R_coup lam2 t0 t t := by
    have hden_pos : t + t0 > 0 := StS_denominator_pos t0 t
       ht0 ht
    have hden_ne : t + t0 <> 0 := ne_of_gt hden_pos
```

```
    have hR0seq_pos : R_base + R_prod + R_mark > 0 :=
       hR_base_pos
    have hseq_ne : R_base + R_prod + R_mark <> 0 := ne_of_gt
        hR0seq_pos
    unfold CBR StS
    have key : forall lam' : R,
    (R_base + R_prod + R_mark + R_coup) * Real.exp (-lam' *
       t) / (t + t0) /
    ((R_base + R_prod + R_mark) * Real.exp (-lam' * t) / (t
       + t0))
    = (R_base + R_prod + R_mark + R_coup) / (R_base + R_prod
        + R_mark) := by
    intro lam'
    have hexp_ne : Real.exp (-lam' * t) <> 0 := ne_of_gt (
       Real.exp_pos _)
    field_simp
    rw [key lam1, key lam2]
```

Significance. Formal basis for the paper's sensitivity-analysis discussion — in the equal-deployment-time special case, the exponential decay terms cancel algebraically, leaving CBR dependent only on the ratio of intrinsic quality terms. Note: this result does *not* extend to the general unequal-time case (explicitly flagged as an open limitation in the paper).

14 CBR Equals 1 Under No Coupling

Statement. When $R_{\text{coup}} = 0$ and $t_{\text{int}} = t_{\text{seq}}$, the Coupling Benefit Ratio equals exactly 1.

$$R_{\text{coup}} = 0,\ t_{\text{int}} = t_{\text{seq}} \implies \text{CBR} = 1 \tag{21}$$

```
    theorem CBR_eq_one_iff_no_coupling
    (R_base R_prod R_mark lam t0 t : R)
    (ht0 : t0 > 0) (ht : t >= 0)
    (hR_base_pos : R_base + R_prod + R_mark > 0) :
    CBR R_base R_prod R_mark 0 lam t0 t t = 1 := by
    have hden : t + t0 > 0 := StS_denominator_pos t0 t ht0
       ht
    have hStS_pos : StS (R_base + R_prod + R_mark) lam t0 t
       > 0 := by
    unfold StS
    apply div_pos
    . exact mul_pos hR_base_pos (Real.exp_pos _)
    . exact hden
    unfold CBR
    rw [add_zero]
    exact div_self (ne_of_gt hStS_pos)
```

Significance. Confirms the paper's stated interpretation that CBR = 1 indicates "no integration advantage" — precisely in the case where there is no coupling value and both deployment paths take equal time, integration provides mathematically zero benefit, exactly as the model predicts.

#	Theorem	Category
1	`StS_denominator_pos`	Well-definedness
2	`StS_well_defined`	Well-definedness
3	`StS_max_value`	StS properties
4	`MarketResponse_pos`	Positivity
5	`TemporalDecay_bounds`	Bounds
6	`trend_half_life`	Decay properties
7	`decay_uniqueness_given_derivative_at_zero`	Decay properties
8	`eta_coup_bounds`	CBR components
9	`decay_rate_estimator_well_defined`	Parameter estimation
10	`StS_strictly_decreasing`	Monotonicity
11	`StS_max_at_zero`	Monotonicity
12	`CBR_geq_one`	CBR main result
13	`CBR_lambda_invariant_equal_time`	CBR sensitivity
14	`CBR_eq_one_iff_no_coupling`	CBR special case

Table 1: Complete list of the fourteen machine-checked theorems.

Summary

Scope Limitations, Stated Honestly

- This verification establishes internal mathematical consistency of the model as defined — it does not constitute empirical validation against real-world FMCG data.
- The sensitivity of CBR to λ under *unequal* deployment times ($t_{\text{int}} \neq t_{\text{seq}}$) has not been formally characterised (Theorem 13 covers only the equal-time case).
- Theorem 7's uniqueness result is proven under an explicit differentiability hypothesis, not derived from continuity and the memoryless functional equation alone; the fully general classical characterisation remains unformalised.